

MOYYNTY, Kazakhstan

WMO: 357930

Lat: 47.217N

Lon: 73.350E

Elev: 585

StdP: 94.49

Time Zone: 6.00 (E06)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.6	-26.0	-32.5	0.2	-27.9	-30.0	0.3	-25.5	14.1	-3.2	10.0	-6.3	0.8	0	0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	33.8	16.5	32.1	16.1	30.5	15.6	18.1	28.2	17.4	27.9	16.7	27.4	3.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.5	11.8	20.2	13.9	10.7	20.2	12.5	9.7	19.7	53.6	27.9	51.3	28.1	49.1	27.5	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.3	7.3	-32.3	37.6	3.0	2.2	-34.5	39.1	-36.2	40.4	-37.9	41.7	-40.1	43.3		
(5)			-32.4	19.8	3.0	1.6	-34.6	21.0	-36.4	21.9	-38.0	22.9	-40.2	24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-13.7	-13.9	-6.1	7.5	14.3	20.6	23.0	20.6	13.9	5.1	-4.1	-10.7	(6)	
(7)		DBStd	14.23	5.96	5.74	5.91	5.25	4.34	4.01	2.99	3.25	4.05	5.19	5.86	6.05	(7)	
(8)		HDD10.0	3268	733	670	499	108	15	1	0	0	12	167	422	642	(8)	
(9)		HDD18.3	5279	992	903	757	326	134	24	4	16	141	411	672	900	(9)	
(10)		CDD10.0	1373	0	0	0	32	147	318	402	329	129	15	0	0	(10)	
(11)		CDD18.3	342	0	0	0	1	8	92	148	86	7	0	0	0	(11)	
(12)		CDH23.3	4406	0	0	0	16	177	1179	1791	1069	166	8	0	0	(12)	
(13)		CDH26.7	1679	0	0	0	2	32	457	761	396	29	1	0	0	(13)	
(14)	Wind	WSAvg	2.5	1.7	2.0	2.4	3.3	3.2	3.0	2.7	2.7	2.4	2.5	2.1	1.7	(14)	
(15)	Precipitation	PrecAvg	325	36	33	34	28	26	25	30	16	10	19	31	36	(15)	
(16)		PrecMax	701	98	73	80	62	95	87	95	48	45	64	68	95	(16)	
(17)		PrecMin	167	5	4	8	0	3	3	4	1	0	2	1	7	(17)	
(18)		PrecStd	131	20	17	17	14	18	19	19	11	12	17	18	18	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.2	0.2	12.3	25.2	29.7	35.1	36.6	35.4	30.1	23.7	11.5	2.9	(19)	
(20)			MCWB	-0.7	-0.7	7.8	12.2	14.4	16.7	17.4	16.7	14.6	11.7	6.6	1.0	(20)	
(21)		2%	DB	-2.3	-1.4	7.6	22.1	27.4	32.7	34.2	33.2	27.2	19.6	7.7	0.3	(21)	
(22)			MCWB	-3.0	-2.1	4.9	10.9	13.6	16.0	16.9	15.8	13.0	9.6	4.7	-0.7	(22)	
(23)		5%	DB	-3.8	-3.0	4.1	19.7	25.5	31.1	32.4	30.8	25.4	16.6	5.3	-1.0	(23)	
(24)			MCWB	-4.4	-3.6	2.4	10.0	12.9	15.7	16.4	15.1	12.5	8.1	3.3	-1.8	(24)	
(25)		10%	DB	-5.4	-4.9	1.6	17.0	23.4	29.4	30.8	29.0	23.2	13.7	2.8	-2.4	(25)	
(26)			MCWB	-5.9	-5.7	0.3	8.7	12.1	15.1	16.1	14.7	11.6	7.0	0.9	-3.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	-0.3	8.7	13.5	16.3	18.8	19.6	18.1	15.2	12.5	7.3	1.1	(27)	
(28)			MCDWB	-0.3	-0.1	11.2	22.3	24.8	29.9	26.7	31.2	27.4	20.4	10.0	1.9	(28)	
(29)		2%	WB	-2.8	-2.0	5.2	12.0	15.0	17.7	18.5	16.9	13.8	10.6	5.0	-0.5	(29)	
(30)			MCDWB	-2.4	-1.4	7.0	19.7	23.7	27.7	27.7	29.1	25.6	17.6	6.6	0.2	(30)	
(31)		5%	WB	-4.4	-3.7	2.6	10.7	13.9	17.0	17.7	16.1	12.8	8.9	3.3	-1.7	(31)	
(32)			MCDWB	-3.9	-3.0	4.2	17.6	22.7	27.3	27.9	27.4	23.9	15.5	5.4	-1.0	(32)	
(33)		10%	WB	-5.9	-5.6	0.5	9.3	13.1	16.1	17.1	15.5	11.9	7.6	1.2	-3.0	(33)	
(34)			MCDWB	-5.5	-4.9	1.5	16.2	21.6	26.5	27.8	26.5	22.2	13.1	2.4	-2.4	(34)	
(35)	Mean Daily Temperature Range			MDBR	9.3	10.1	9.2	12.3	13.4	13.6	13.3	13.9	14.3	11.1	8.6	8.7	(35)
(36)		5% DB	MCDBR	8.7	7.9	10.1	16.0	16.6	15.7	15.4	16.5	17.0	16.1	10.4	8.5	(36)	
(37)			MCWBR	8.4	7.3	7.3	7.6	6.8	5.5	5.4	6.2	8.0	8.8	7.6	7.6	(37)	
(38)			MCDBR	8.5	7.9	9.3	14.1	13.8	14.0	13.1	14.3	15.6	14.3	9.6	7.6	(38)	
(39)		5% WB	MCWBR	8.3	7.4	7.0	7.4	6.4	5.5	4.8	5.7	7.6	8.4	7.3	7.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.299	0.329	0.379	0.419	0.427	0.420	0.432	0.387	0.343	0.343	0.329	0.301	(40)	
(41)		taud		2.075	2.008	2.050	2.080	2.113	2.197	2.176	2.292	2.375	2.295	2.207	2.132	(41)	
(42)		Ebn at Noon		750	806	825	830	839	846	828	855	865	796	715	698	(42)	
(43)		Edh at Noon		93	124	141	151	151	140	141	121	101	93	82	79	(43)	
(44)	All-Sky Solar Radiation	RadAvg		1.72	2.79	4.03	5.34	6.66	7.01	6.63	6.21	4.85	2.92	1.59	1.23	(44)	
(45)		RadStd		0.22	0.33	0.33	0.43	0.39	0.39	0.45	0.36	0.28	0.26	0.17	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page